High-Frequency Characterization of C_{OSS} Loss and Dynamic R_{dsON} in GaN

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***Abstract*—Gallium nitride (GaN) transistors are enabling higher power density in next-generation power electronics, but unmodeled loss mechanisms such as dynamic on-resistance (R_{dsON}) and output capacitance (C_{oss}) losses can limit their performance, especially at multi-megahertz frequencies. This work presents a comprehensive experimental characterization of R_{dsON} and C_{oss} losses for commercially available GaN devices under class-E type waveforms such as used in switched-mode RF power amplifiers and dc-dc converters. We measure device operation at duty cycles of 25% and 50%, switching frequencies up to 30 MHz, peak voltages from 200 V to 400 V, and case temperatures up to 80 °C. The resulting data and trends enable more accurate device modeling and selection, supporting optimized high-efficiency high-frequency converter designs.**

***Index Terms*—GaN Power Devices,**

I. INTRODUCTION

Soft-switched power converters that operate at high frequencies offer several advantages, including improved performance, smaller energy storage components, and reduced overall size [1]. Gallium Nitride (GaN) transistors have drawn significant interest for enabling such converters due to their fast switching capability, low on-resistance, and reduced switching losses [2]. However, these devices exhibit loss mechanisms such as increased ON-state resistance (dynamic R_{dsON}) for high-frequency switching caused by electron trapping within defects in the material [3], and output capacitance (C_{oss}) losses during the OFF-state resulting from the charging and discharging of the output capacitor every switching cycle [4]. Omitting these losses from the design calculations can lead to lower than expected experimental efficiency, especially at high frequencies, and can significantly compromise converter design optimization [5].

Considerable work has been done to experimentally characterize these losses using a range of measurement techniques [3], [4], [6]–[13]. Dynamic R_{dsON} behavior has been well researched and characterized for sub-megahertz operation [6], [7], with [3] extending these measurements to 3 MHz. C_{oss} loss measurements for 5–35 MHz and multiple waveform shapes using the Sawyer–Tower method are reported in [4]. However, many of the devices in [3], [4] are now obsolete, highlighting the need for updated data on currently available parts. Data from more recent work on automating the Sawyer–Tower method [13] is limited to a single GaN device and 10 MHz. Other studies [10]–[12] present alternative C_{oss} measurement methods but do not provide data for commercial devices that designers can readily apply.

Previous research has shown that C_{oss} losses vary substantially among devices and are influenced by multiple factors, including peak device voltage, dv/dt, device temperature, and operating frequency [4], [8]. Most published data, however, is limited to narrow operating ranges, often at a single duty cycle, frequency, temperature, and voltage. This combination of loss dependence on multiple factors and limited experimental data makes it difficult to generalize loss behavior and highlights the need for comprehensive, up-to-date experimental data to improve loss modeling and enable higher-efficiency designs.

In this work, we present an extensive experimental study of C_{oss} losses and dynamic R_{dsON} for commercial 650 V GaN devices under waveforms representative of those in a Class-E power converter. Measurements are performed across duty cycles of 25 % and 50 %, switching frequencies up to 30 MHz, peak switch voltages from 200 V to 400 V, and case temperatures up to 80 °C. To examine intra-family scaling, devices from the same product family with different die sizes are also evaluated. In addition, a 200-V GaN device is characterized to illustrate how losses scale relative to higher-voltage-rated devices.

II. MEASUREMENT METHOD

The measurement circuit of [3], originally developed for dynamic R_{dsON} measurement in GaN transistors at 3 MHz, is adapted here to measure both dynamic R_{dsON} and C_{oss} losses at frequencies up to 30 MHz. The simplified schematic shown in Fig. 1 consists of a DC power supply, a two-stage high-frequency filter, input inductor (L_R), shunt capacitor (C_R), and two identical devices under test (Q1, Q2). The input inductor resonates with the shunt capacitor in parallel with device capacitances, producing a half-sine pulse of voltage (V_{PK}) at the switch node, similar to the switch node waveform in a Class-E converter, as shown in Fig. 2.

The values of V_{DC}, L_R, and C_R are adjusted to achieve the desired V_{PK} for a given frequency and duty cycle while maintaining zero-voltage switching. For each test point, the circuit is configured and tested in three different ways to extract both C_{oss} loss and dynamic R_{dsON}. The input power to this unloaded resonant circuit corresponds to the total loss in the circuit and is measured using precision multimeters for

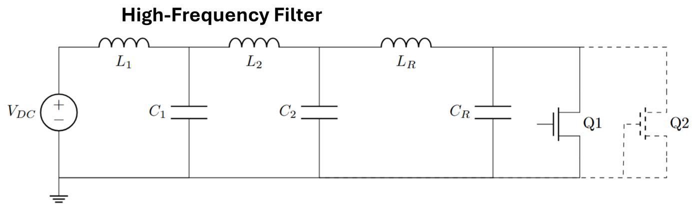


Fig. 1: Simplified circuit diagram of the measurement circuit along with a two-stage high-frequency filter adapted from [3]. Q2 connections are shown dashed to illustrate the various test configurations employed.

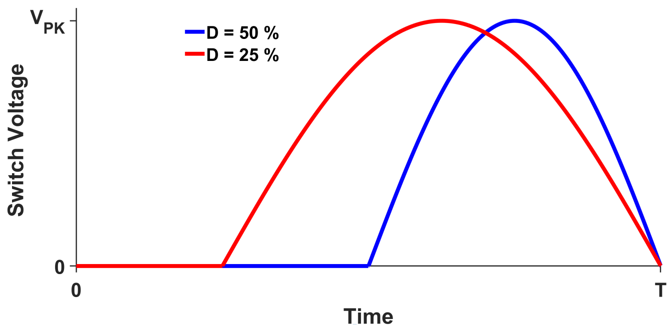


Fig. 2: Switch node voltage waveforms with zero-voltage turn-on for different duty cycles.

each configuration. High-Q capacitors are used for the shunt capacitor branch, and their losses are omitted in this study.

Configuration 1 (Q2 Always OFF): Both identical devices are mounted on the circuit and the gate of the second device (Q2) is grounded so it remains OFF. The measured input power includes inductor loss (P_{L_R}), conduction loss ($P_{\text{cond}} = I^2_{\text{sw}_{\text{rms}}} R_{\text{dynamic,ON}}$), C_{oss} losses of Q1 (P_{Coss}), C_{oss} loss of Q2 (P_{Coss}), and other losses such as PCB dielectric loss and switching loss (P_{other}).

$$P_{\text{in},1} = P_{L_R} + P_{\text{cond}} + 2P_{\text{Coss}} + P_{\text{other}}$$

Configuration 2 (Q1 and Q2 Current Sharing): Both identical devices are mounted on the circuit and driven with the same gate signal to achieve equal current sharing. Q1 now carries half the current compared to Configuration 1. Total conduction loss in this case is $2\,(\frac{I_{\text{sw}_{\text{rms}}}}{2})^2 R_{\text{dynamic,ON}}$. The total input power is :

$$P_{\text{in},2} = P_{L_R} + \frac{P_{\text{cond}}}{2} + 2P_{\text{Coss}} + P_{\text{other}}$$

Subtracting input power for Configuration 2 from Configuration 1 yields the single-device conduction loss, which can then be used to determine the dynamic on-resistance:

$$P_{\text{cond}} \;=\; 2\,(P_{\text{in},1} \;-\; P_{\text{in},2}) = I^2_{\text{sw}_{\text{rms}}} R_{\text{dynamic,ON}}$$

Configuration 3 (Q2 Removed): Q2 is removed from the circuit, and C_R is augmented with an additional linear capacitor $C_{\text{oss,eq}}$ that replaces the nonlinear C_{oss} of Q2 so the total shunt capacitance at the switch node matches Configuration 1. GaN devices have a strongly voltage-dependent C_{oss}, so $C_{\text{oss,eq}}$ is selected per operating point using the charge-equivalent definition from [14][1]. Matching the total capacitance at the switch node in both configurations ensures that the resonant current remains the same, keeping all current-dependent losses unchanged. The input power in this case is:

$$P_{\text{in},3} = P_{L_R} + P_{\text{cond}} + P_{\text{Coss}} + P_{\text{other}}$$

Taking the difference of the measured powers from Configuration 1 and Configuration 3 yields the single-device C_{oss} loss:

$$P_{\text{Coss}} \;=\; P_{\text{in},1} \;-\; P_{\text{in},3}$$

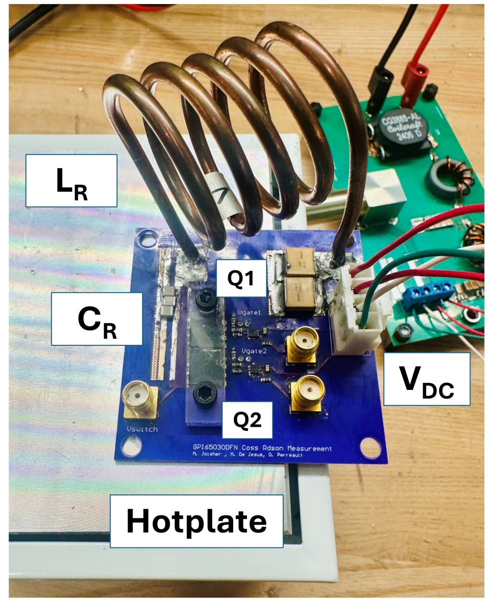


Fig. 3: Experimental test setup for Configuration 1, with the board mounted on a hotplate and DC power supplied through a separate high-frequency input filter board

This conduction loss measurement approach differs from that in [3], where conduction loss is back-calculated by estimating other losses in the circuit. As the operating frequency increases to 10 MHz and above, accurately estimating inductor loss and other system losses becomes difficult. The three-configuration method used here avoids that issue by keeping the current in the circuit constant across configurations and using power differences to separate C_{oss} and conduction losses.

III. Test Setup

The measurement setup includes a DC power supply, two multimeters for monitoring input current and voltage, an input filter board, a high-frequency GaN measurement board, an

[1]Additional tuning may be required to match the circulating current between configurations.

oscilloscope, and a hot-plate. The input filter board is placed between the supply, multimeters, and the GaN board to prevent high-frequency switching noise from corrupting the DC input power measurements. A two-stage high-frequency input filter is used, as shown in Fig. 1. The cutoff frequencies of the two stages are selected to provide sufficient attenuation over the operating frequency range. The characteristic impedances of the filter sections are chosen such that the converter's minimum input impedance is significantly higher than the filter's output impedance, ensuring stability and minimizing filter–converter interaction [15]. The high-frequency measurement board integrates the resonant inductor L_R, resonant capacitor C_R, and the DUTs (Q1 and Q2), along with their associated gate-drive circuitry. The resonant inductor is custom-fabricated from copper tubing to achieve low loss at high frequencies. However, at 27 MHz the required inductance is only a few tens of nHs—approximately a single turn. To realize this inductance while maintaining low loss, the design is transitioned to a single-turn copper-foil inductor. For C_R, high-Q AYX1111 capacitors are paralleled to achieve the target capacitance and meet the required current rating. Both ground-referenced GaN devices are driven using the LMG1025 gate driver. The high-frequency GaN measurement board is fabricated on Rogers 4350B substrate to minimize parasitic capacitance and associated losses thus ensuring that device loss remains the dominant loss mechanism.

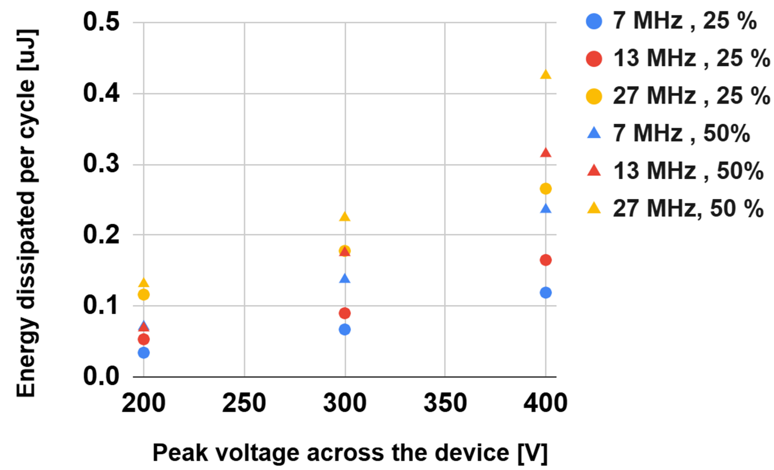
The measurement board is mounted on a hot plate as shown in Fig. 3 to control the case temperature T_c to study the temperature dependence of C_{oss} loss. For bottom-side cooled packages, the dominant case-to-sink thermal path is through the PCB; dense thermal vias and large copper pours are used to minimize the case-to-sink thermal resistance. A plastic clamping block is placed on top of the devices to apply uniform mechanical pressure and further improve thermal contact.

IV. Experimental Results

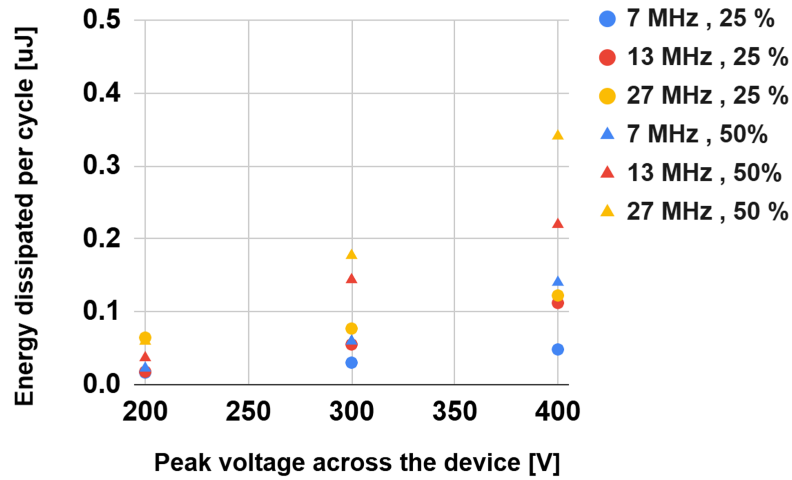
We present results for commercial 650 V GaN devices from the Infineon G5 series, GaN Systems, and GaNPower. Table I summarizes the devices selected for evaluation. Devices with comparable current ratings and on-resistance are chosen to enable cross-manufacturer comparison. In addition, within each manufacturer's device family, devices with different die sizes are included to examine how loss scales with die area. For each $\sim$ 100 mΩ on-resistance device, measurements are performed at two duty cycles (25 % and 50 %), three switching frequencies (7 MHz, 13 MHz, and 27 MHz), three

TABLE I: Summary of Device Specifications

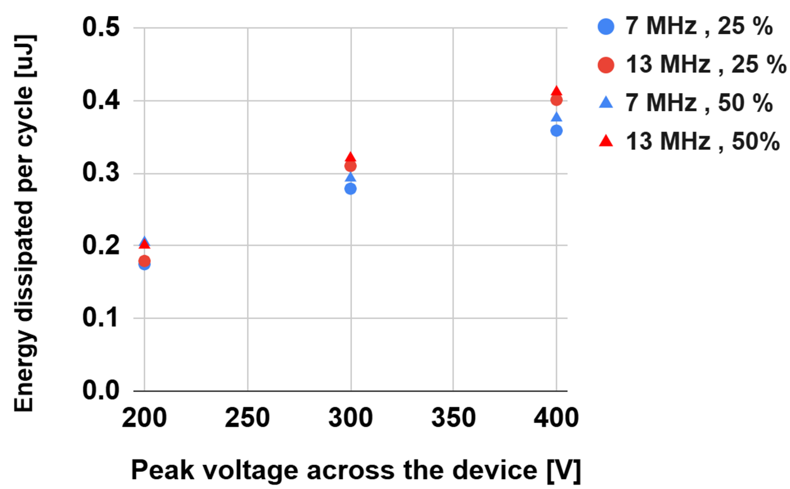
Device	Package	$R_{\text{DS(on)}}$ (mΩ)	C_{oss} (pF)
GS66504B	DFN 5×6	100	31
GS66508B	DFN 8×8	50	65
IGT65R140D2	PG-HSOF-8	140	22
IGT65R045D2	PG-HSOF-8	45	72
GPI65015DFN	DFN 8×8	85	34



(a) Infineon IGT65R140D2

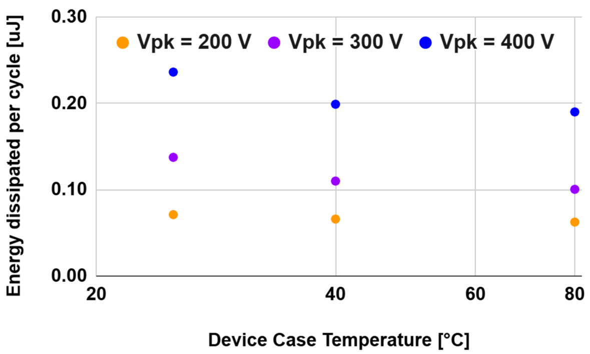


(b) GaN Systems GS66504B

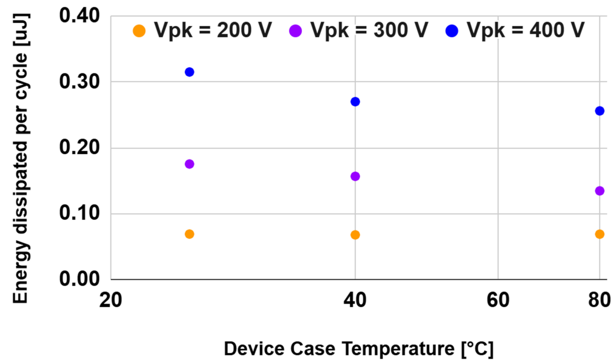


(c) GaNPower GPI65015DFN

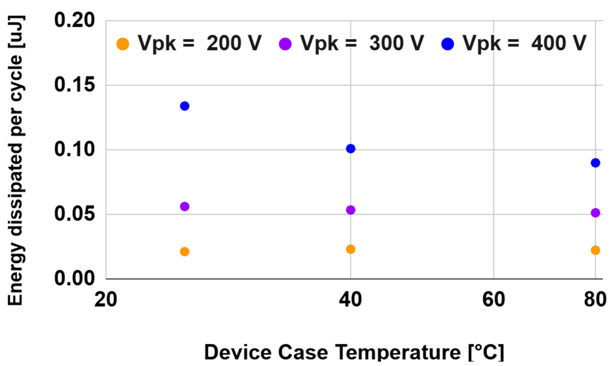
Fig. 4: C_{oss} energy dissipated per cycle in each device for a given switching frequency and duty cycle at 25°C case temperature.



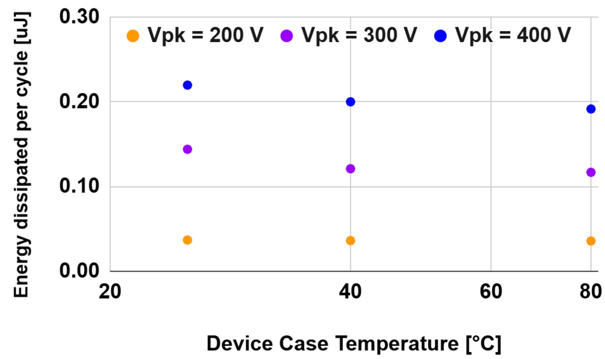
(a) IGT65R140D2, 7 MHz



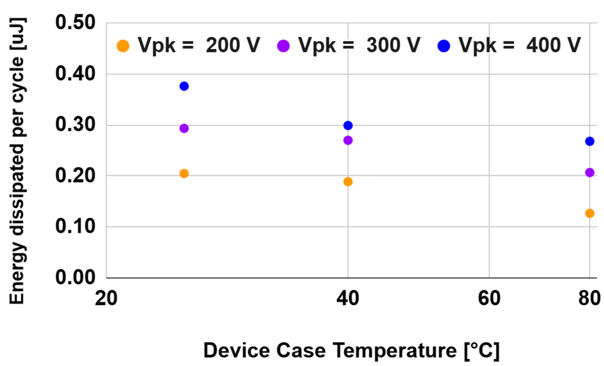
(b) IGT65R140D2, 13 MHz



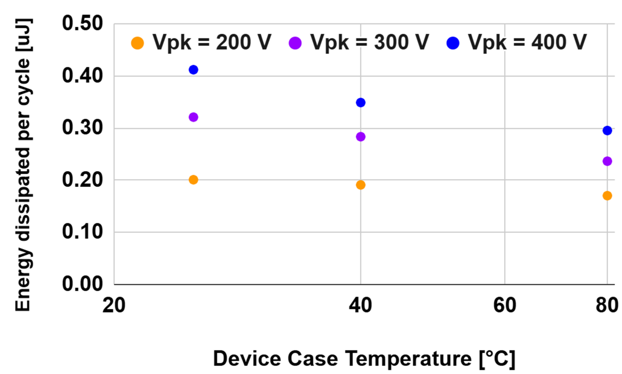
(c) GS66504B, 7 MHz



(d) GS66504B, 13 MHz



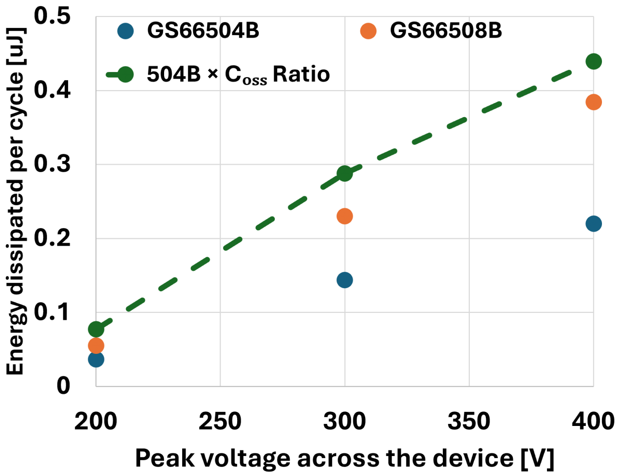
(e) GPI65015DFN, 7 MHz



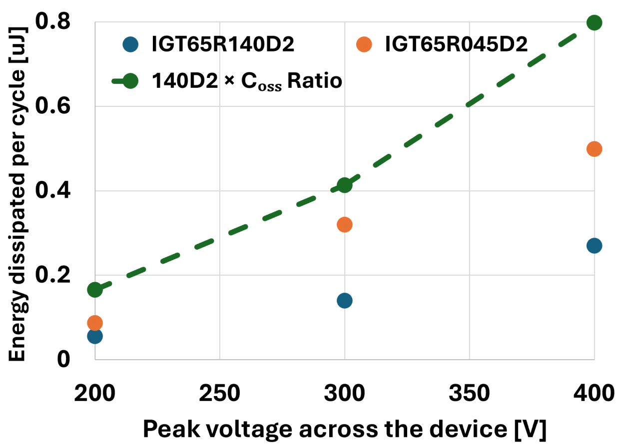
(f) GPI65015DFN, 13 MHz

Fig. 5: Variation of C_{oss} loss with case temperature at different switching frequencies, measured at 50% duty cycle.

peak drain–source voltages ($V_{\text{DS}} = 200\text{ V},\ 300\text{ V},\ 400\text{ V}$), and three device case temperatures (25 °C, 40 °C, and 80 °C). Fig. 4 shows the energy dissipated per cycle in charging and discharging the output capacitance for different combinations of switching frequency and duty cycle at case temperature of 25 °C. The data show that C_{oss} loss increases exponentially with device voltage and also rises with frequency for a given duty cycle. The rise in C_{oss} loss with voltage has been linked to substrate-related loss mechanisms, where higher electric fields drive greater displacement current through the GaN/Si stack and increases with frequency are associated with trap-related mechanisms, as shorter switching periods limit the time available for charge detrapping [8]. For a given frequency and voltage, the dependence on duty cycle, and therefore on the dv/dt, varies across devices. For IGT65R140D2 and GS66504B, C_{oss} loss exhibits strong dependence on duty cycle, indicating sensitivity to dv/dt. Particularly in GS66504B, at higher frequencies the loss at 25% duty for the same peak voltage is less than half of that at 50% duty. This provides an important insight for designers that operating at 25% duty instead of 50% can yield substantial C_{oss} loss reduction. In contrast, for the GPI65015DFN C_{oss} loss appears to be



(a) GS66504B and GS66508B



(b) IGT65R140D2 and IGT65R045D2

Fig. 6: C_{oss} energy dissipated per cycle for devices within the same family at 13 MHz, 50% duty, and a case temperature of 25°C.

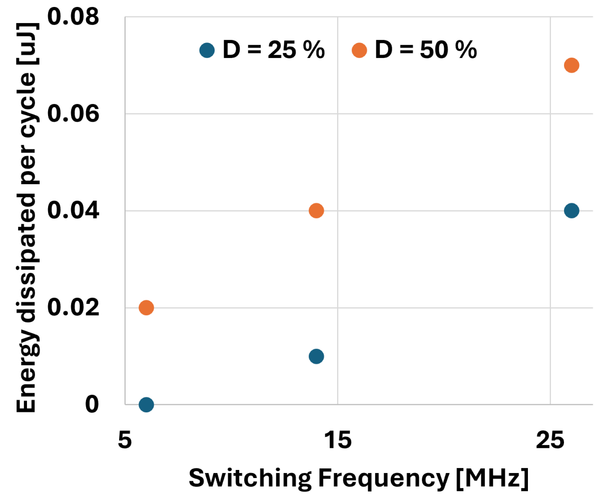


Fig. 7: C_{oss} energy dissipated per cycle for EPC 2307 for different on-duties under a 100 V peak Class-E waveform at a case temperature of 25°C.

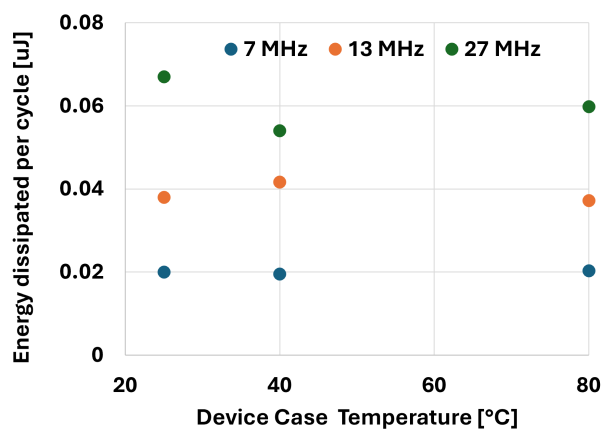


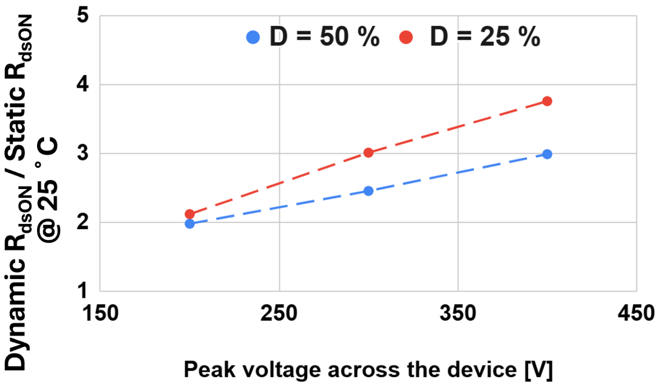
Fig. 8: Variation of C_{oss} loss with case temperature at different switching frequencies, measured at 50% duty cycle and 100 V_{PK}

primarily dictated by peak voltage, with minimal dependence on frequency and dv/dt.

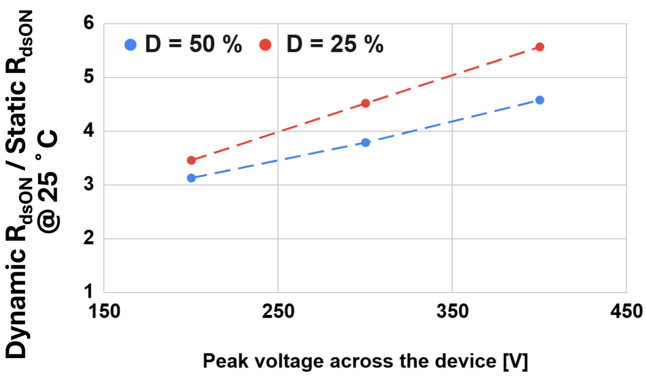
Fig. 5 shows the variation of C_{oss} loss with case temperature for GS66504B, IGT65R140D2, and GPI65015DFN at different voltages and frequencies. The results indicate a strong temperature dependence, with C_{oss} loss decreasing as case temperature increases. This observation is consistent with the findings in [8], which attribute the reduction to increased carrier energy at higher temperatures, enabling trapped carriers to escape material defects more easily.

Next, to evaluate how loss scales within device families, we characterized ~ 50 mΩ devices from the same device families. Fig. 6a presents the C_{oss} loss results for the GS66504B and GS66508B, along with the predicted loss for the GS66508B obtained by scaling the GS66504B data using the ratio of device C_{oss}. Consistent with prior findings [4], the GaN Systems devices exhibit strong intra-family scaling: loss predicted by C_{oss} ratios is higher than the measured loss, indicating that this scaling relationship provides a reliable upper bound for C_{oss} loss. Similar behavior is observed in Infineon's G5 series, as shown in Fig. 6b.

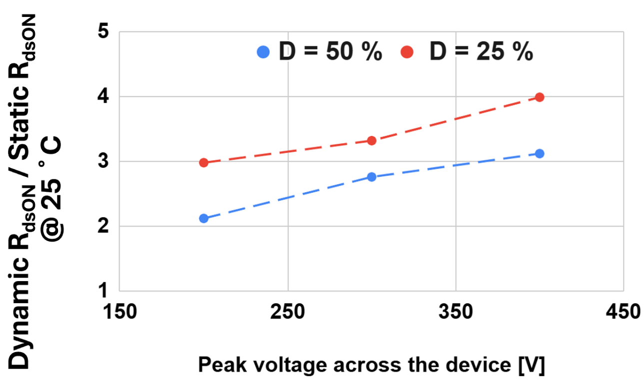
As established in prior sections, commercial 650 V GaN devices exhibit considerable C_{oss} loss under HF/VHF switching conditions, making them a key limitation in high-frequency power-stage efficiency. To explore device options that may mitigate this constraint, we measured a low-voltage EPC device. Fig. 7 presents C_{oss} loss data for the EPC2307 tested with a peak drain voltage of 100 V. Given the strong dependence of C_{oss} loss on applied drain voltage, such low-voltage devices offer an attractive pathway for extending operation into higher-frequency regimes. The EPC2307 exhibits similar frequency, duty-cycle-dependant trends as the 650 V devices shown above, but with substantially lower absolute



(a) IGT65R140D2



(b) GS66504B



(c) GPI65015DFN

Fig. 9: Dynamic R_{dsON} measured at 7 MHz and 25 °C case temperature for commercial 650 V GaN devices.

loss due to the reduced voltage across the switch. Fig. 8 shows the variation of C_{oss} loss with case temperature at different switching frequencies. Unlike the 650 V devices, no strong temperature trend is observed because the absolute loss in this low-voltage device is very small and can be influenced by measurement noise and other system effects. However, the measured C_{oss} loss seems to remain approximately constant and even slightly decreases in some cases as the device case temperature increases.

Next, we characterized the dynamic R_{dsON} for ~ 100 mΩ devices from Table I. Fig. 9 shows the dynamic R_{dsON} at 7 MHz, normalized to the datasheet value at 25 °C, for different duty cycles. For the same peak voltage across the device, the results indicate higher dynamic R_{dsON} at lower duty cycles. This is consistent with past findings at lower frequencies [7] and is likely because reducing the duty ratio shortens the ON time and limits the time available for electron detrapping, leaving more electrons trapped. Dynamic R_{dsON} is also observed to increase with higher switching voltages, which has been linked to stronger electric fields promoting increased electron trapping [3]. Across all devices tested, the dynamic R_{dsON} exceeded twice the datasheet value, and in some cases reached up to six times the datasheet value at higher voltages and lower duty cycles. It should be noted that the GS66504B device operating at 25% duty exhibited the lowest C_{oss} loss among the devices in Table I, yet it also showed a higher dynamic R_{dsON} factor compared to the others. This contrast illustrates an important design tradeoff. In high-frequency, high-power applications where both conduction loss and C_{oss} loss contribute significantly to overall efficiency, device selection must balance these two mechanisms. In such regimes, the optimal device choice must be determined through optimization that captures the interaction between C_{oss} loss, dynamic R_{dsON}, and the converter operating conditions. This underscores the importance of having such detailed experimental data to support informed device selection and to enable accurately modeling and accounting for these losses in converter designs.

V. Conclusion

This work presents a measurement method and comprehensive experimental data for C_{oss} loss and dynamic R_{dsON} in commercial GaN devices across a wide range of voltages, frequencies, duty cycles, and temperatures. We evaluate five 650 V GaN devices from multiple manufacturers and establish intra-family scaling trends that enable first-order estimation of C_{oss} loss without requiring full characterization of every device in the device family. The temperature dependence of C_{oss} loss is also examined, and the measured loss is observed to decrease with increasing case temperature. In addition, the C_{oss} loss of a 200 V GaN device is characterized and shown to be significantly lower than that of 650 V devices, suggesting that lower-voltage-rated devices provide a promising path for achieving high efficiency in VHF applications. Dynamic R_{dsON} is measured for several devices and is found to be significantly higher than the corresponding static value. Overall, the results illustrate how these loss mechanisms vary across device technologies and operating conditions, providing data and insights that support device selection, improve modeling accuracy, and enable higher-efficiency high-frequency designs.

Appendix A
Maintaing Zero-Voltage Switching

To accurately measure C_{oss} loss using the proposed measurement method, it is essential to ensure that the device operates with zero-voltage switching (ZVS). Without ZVS, the device experiences hard switching, making it difficult to separate C_{oss} loss from additional switching loss. Prior work [3] employed a closed-loop ZVS detection circuit to automatically adjust the on-time and switching frequency to maintain ZVS; however, this approach was demonstrated at a lower frequency (3 MHz). When attempting to operate near 30 MHz, the propagation delays and finite response time of the control circuitry become significant, making closed-loop ZVS enforcement difficult to achieve reliably.

Hence, we run this in open loop and manually tune the circuit to maintain ZVS. For a given desired frequency and duty cycle, we first select L_R and C_R while accounting for device capacitance using the charge-equivalent $C_{\mathrm{oss,eq}}$ from [14] evaluated at 200 V_{PK} across the device. As the operating frequency approaches 30 MHz, the required C_R becomes only a few pFs and often negligible compared to the device C_{oss}, making the operating point highly sensitive to the nonlinear, voltage-dependent nature of C_{oss}. To maintain ZVS up to 400 V_{PK} across the device, we incrementally adjust the switching frequency and use waveform inspection together with a minimum-input-power criterion to identify the ZVS condition. If the required frequency adjustment for higher voltages across the device becomes too large relative to the intended operating point, we retune L_R and C_R to re-establish ZVS at our desired frequency.

Appendix B
Data Acquisition Automation

Testing a single GaN device across three frequencies, two duty cycles, three case temperatures, and three voltages yields 54 operating points. With three measurement configurations required per point for the proposed loss–extraction method, each device requires 162 test runs. Each frequency–duty combination also requires retuning of L_R and C_R, and small frequency adjustments may be necessary at different voltages to account for the nonlinear voltage dependence of C_{oss} and maintain ZVS, as described in Section A. These tuning requirements make manual testing slow, error-prone, and difficult to perform with the consistency needed for reliable measurement and data logging. To address this challenge, we developed an automated SCPI-based test platform that assists with frequency tuning, data logging, and operating-point tracking.

As described in Section A, the operating point must be tuned to maintain ZVS, particularly at higher voltages where the nonlinear behavior of C_{oss} becomes significant. In the automated workflow, we begin by manually setting L_R and C_R for the desired frequency–duty combination and choosing the appropriate measurement configuration from Section II. Once a test is initiated, the software records the input dc current and incrementally adjusts the switching frequency within a small range to locate the minimum-input-power point that indicates ZVS. After each run, the dc voltage, tuned frequency, inductor current, and switch current are logged to the cloud and post processed to extract the losses. An oscilloscope screenshot is also saved to verify data integrity in the rare case of logging anomalies. Because the ZVS tuning required at a given operating point remains fairly consistent across temperatures and configurations, the frequency adjustments identified in one setup can be reused for the remaining test points. While users still manually set L_R and C_R to establish the initial operating point, the platform substantially improves measurement throughput, repeatability, and overall testing accuracy. The test software used in this work is available at https://github.com/MatthewASC20/GaN-FET-Characterisation.

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